From Pixels to Patches

Pooling Strategies for Earth Embeddings

Main result: simple distributional pooling improves spatial generalization for fixed embedding products.

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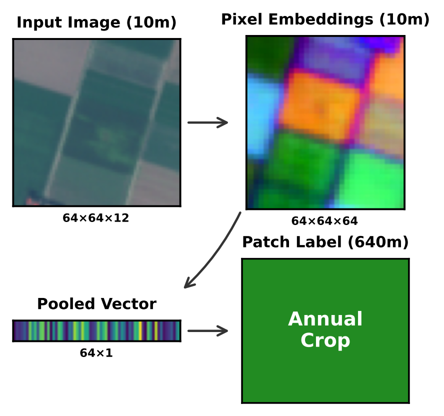
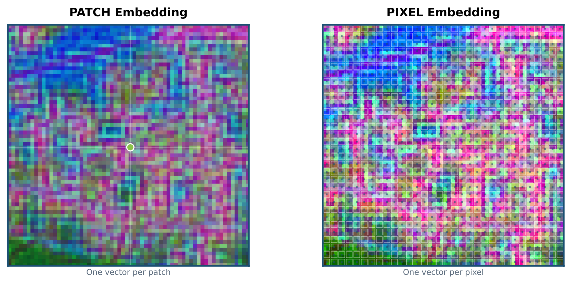


Pooling is the bottleneck

- Pixel-level GFMs output one embedding *per pixel*.
- Downstream labels live at the **patch / region** level.
- A EuroSAT patch contains **4,096** embeddings that must be collapsed to one vector.

Why this matters

Mean pooling can lose **8.8 pp** under geographic shift.



EuroSAT-Embed benchmark

81,000 pixel-embedding GeoTIFFs aligned to EuroSAT (27k patches, 10 classes) for **reproducible pooling research** without rerunning GFM inference.

AlphaEarth	**OlmoEarth-Nano**	**Tessera**
64-d `int8`	128-d `float32`	128-d `uint8`
3.3 GB	40 GB	11 GB

3 encoders → 13 pools → 2 probes → 2 splits

156 experiments random vs. spatial splits kNN + linear probes

What we benchmark

Mean

$1\times$ dim

baseline

Mean+Max

$2\times$ dim

strong balanced option

Stats

$4\times$ dim

min / max / μ / σ

Covariance

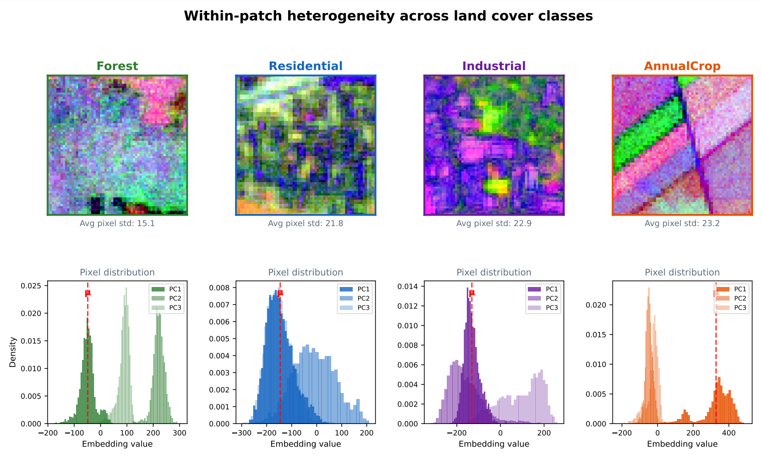
$D(D+1)/2$ dim

second-order structure

13 total methods: 11 training-free pools plus PCA and BoVW.

Mean misses within-patch structure

- Homogeneous classes cluster tightly.
- Heterogeneous classes show broad distributions.
- Mean pooling keeps the center and discards the spread.



A residential patch and an industrial patch can share a similar mean while differing sharply in variance.

Distributional statistics improve spatial generalization

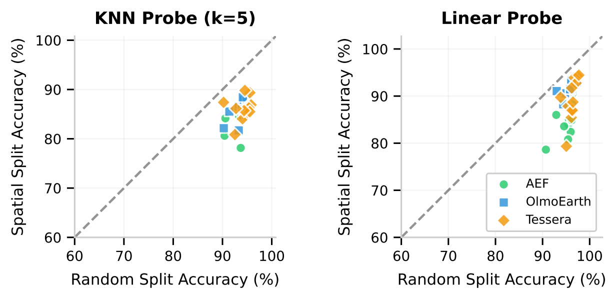
Spatial linear probe average

Mean	87.3
Mean+Max	91.8
Covariance	92.8
Stats	**93.0**

Key result

Stats gains **+5.7 pp** over Mean on spatial splits.

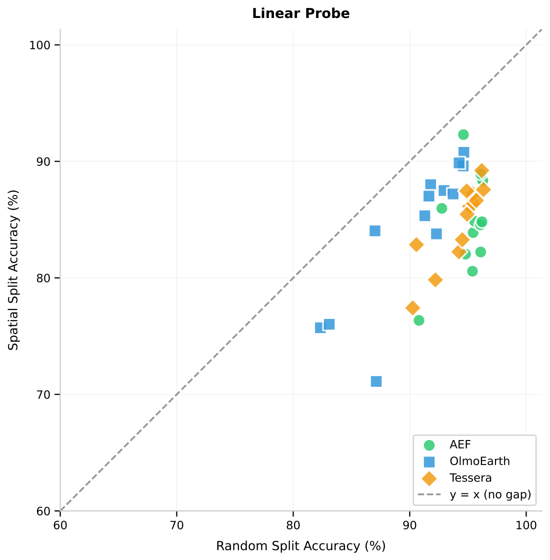
On random splits, the gap shrinks to ~**1 pp**.



Closer to diagonal = smaller random→spatial gap

Spatial evaluation is the honest test.

Accuracy and robustness are aligned



Average gap (linear probe)

Mean	**8.8 pp**
Mean+Max	4.6 pp
Covariance	4.3 pp
Stats	3.8 pp

- No accuracy–robustness tradeoff.
- Best spatial methods also have the smallest gaps.
- **Stats**, **Covariance**, and **Mean+Max** all dominate Mean.

Practical recommendation

Mean	Mean+Max	Stats	Covariance
$1\times$ dim	$2\times$ dim	$4\times$ dim	$D(D{+}1)/2$ dim
cheapest baseline	strong balanced option	best default	peak accuracy

Across embedding products

The recommendation is stable across AlphaEarth, OlmoEarth, and Tessera, though the size of the gain varies by encoder.

Takeaways and scope

Takeaways

- Pooling is **not** a benign preprocessing step for fixed pixel-level embedding products.
- Simple distributional summaries materially improve **spatial** generalization.
- **Stats** is the best default; **Covariance** is strongest when dimensionality is less constrained.

Scope

- Single benchmark: EuroSAT
- Three released embedding products
- Post-hoc pooling only; no encoder access

Questions?

Code

Dataset